NOTATION · CHEAT SHEET

Propositional Logic Notation at a Glance

The marks from the notation cards across the propositional-logic pages: how each is read and what it means. Every group links to the full card on its page.

SYNTAX

$P, Q, R \cdot P_1, P_2$

PROPOSITION LETTERS
Capital letters name atomic propositions; subscripts extend the supply.

$\neg \ \wedge \ \vee \ \rightarrow \leftrightarrow$

NOT; AND; OR; IMPLIES; IF AND ONLY IF
The five connectives: negation, conjunction, disjunction, implication, biconditional.

φ, ψ

PHI AND PSI
Placeholders for whole formulas in the formation rules.

$(\varphi \wedge \psi)$

PARENTHESISED CONJUNCTION
Every binary connective brings its own parentheses; no precedence at this level.

IMPLICATION

$p \rightarrow q$

IF P THEN Q; P IMPLIES Q
The conditional connective: antecedent p, consequent q.

$p \Rightarrow q$

P IMPLIES Q, IN THE META SENSE
The double shaft claims the implication holds in every case; older texts write $p \supset q$.

$p \rightarrow q$ in words

THE ENGLISH PHRASINGS OF ONE ARROW
"if p, then q"; "p is sufficient for q"; "q is necessary for p"; "p only if q".

EQUIVALENCES

$P \equiv Q$

P IS LOGICALLY EQUIVALENT TO Q
A claim about formulas: P and Q agree in every case.

$P \Leftrightarrow Q$

P IF AND ONLY IF Q, IN THE META SENSE
The double-shafted spelling of the same claim.

$P = Q$

P EQUALS Q
Boolean algebra's spelling, where formulas denote values in $\{0, 1\}$.

TAUTOLOGY

$\top$

TOP; VERUM; THE TRUE CONSTANT
Truth itself: a formula true under every assignment.

$\vdash P$

TURNSTILE P; P IS PROVABLE
The single turnstile: a derivation of P exists in the proof system.

$\models P$

DOUBLE TURNSTILE P; P IS VALID
The double turnstile: P is true under every assignment.

$P \equiv \top$

P IS EQUIVALENT TO THE TRUE CONSTANT
The third way to say that P is a tautology.

CONTRADICTION

$\bot$

BOTTOM; FALSUM; THE FALSE CONSTANT
Falsehood itself: a formula false under every assignment.

$\vdash \neg P$

THE NEGATION OF P IS PROVABLE
The proof-side way to declare P a contradiction.

$P \equiv \bot$

P IS EQUIVALENT TO THE FALSE CONSTANT
The equivalence spelling of "contradiction".

SAME MARKS, DIFFERENT JOB

$\rightarrow$ vs $\Rightarrow$	A connective inside a formula; a claim about formulas.
$\leftrightarrow$ vs $\equiv$	The biconditional connective; logical equivalence in the metalanguage.
$\vdash$ vs $\models$	Provable by the rules; true in every case.
$\supset$	Implication in older logic texts; superset in set theory.
$\bot$	The false constant here; perpendicular in geometry.